Gayatri Vidya Parishad College of Engineering Autonomous

Data Science | Course Code: DS301

Applied Statistics & Data Science - Laboratory Manual & Practical Guide

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Syllabus Focus: DS301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Probability Distributions & Sampling Theory

" Syllabus Scope: Discrete and continuous random variables, Normal, Binomial, Poisson, Exponential distri

Unit 2: Parametric Hypothesis Testing

" Syllabus Scope: Null & alternative hypotheses, Type I and Type II errors, p-value interpretation, Z-tes

DS301: Applied Statistics & Data Science - Advanced Topics & Assessment

Academic Track: Data Science - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Categorical Analysis & Chi-Square Tests

" Goodness-of-fit tests, Test of independence in contingency tables, Contingency coefficients, Degrees of free

Unit 4: Analysis of Variance ANOVA

" One-way and Two-way ANOVA, F-distribution, Total sum of squares decomposition, Post-hoc tests Tukey HSD

Unit 5: Correlation & Multivariate Regression

" Pearson and Spearman rank correlation, Multiple linear regression, Multicollinearity, Variance Inflation Fac

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Exploratory Data Analysis, Outlier Detection, and Missing Value Imputation in Python Pandas
Lab 2: Parametric Hypothesis Testing Z-test, T-test, Paired T-test on Real-World Datasets
Lab 3: One-Way and Two-Way ANOVA Execution with F-Distribution Significance Testing
Lab 4: Multiple Regression Modeling with VIF Multicollinearity Diagnostics and Residual Plotting

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Douglas C. Montgomery - Applied Statistics and Probability for Engineers Wiley 7th Ed
* Gareth James et al. - An Introduction to Statistical Learning Springer

Authorized NoteVault Educational Document " Verified for Registered University Students